

EE3900 Linear Systems and Signal Processing, Exam 3

Max. Marks: 22.5. **Time:** 1 hour 15 minutes.

- You are welcome to answer all questions.
- The maximum marks will be capped at 22.5. That is, if your answers are evaluated to x marks then your score will be $\min\{22.5, x\}$.

Instructions

- Please **write your serial number and roll number** prominently in the first page of the answer sheet. Please see the [attendance sheet](#) to know your serial number.
- No breaks during the exam. No laptops, calculators, mobile devices etc. allowed.
- Each step in your answer must be supported by clear mathematical reasoning. For questions requiring you to provide proofs, you must include a precise and rigorous proof.
- **Unless otherwise stated, any result stated or proved in the lectures can be used by stating it clearly, no need to prove them in your answer. Any other result must be proved before using them in your answer.**

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1. (5 marks) Find the Fourier transform of the following distribution T , where III denotes the Dirac comb,

$$\frac{d(\text{III}(t-3))}{dt} \times \exp(-t^2).$$

You must express the distribution $\mathcal{F}T$ as a function of the variable s .

2. (5 marks) Let ψ be a Schwartz function and T be a tempered distribution. Using the properties of the Fourier transforms of Schwartz signals, prove that

$$\mathcal{F}^{-1}(\psi T) = \mathcal{F}^{-1}(\psi) * \mathcal{F}^{-1}(T).$$

Clearly explain all the steps in the proof.

3. (5 marks) Derive the value of the integral

$$\int_{x=-\infty}^{+\infty} \delta'(x) dx.$$

Hint: You can not view this as the distribution δ' acting on a Schwartz function, since 1 is not Schwartz. Instead, take the alternative viewpoint where δ is thought of as the limit of a sequence of functions f_ϵ .

Not a part of this exam (you don't have to solve this): Can you find $\int_{-\infty}^{+\infty} \delta''(x) dx$?

4. (5 marks) You must either prove or disprove the following claim. Here, f and g are signals with corresponding distributions denoted as T_f and T_g , respectively.

Claim: The distributions T_f and T_g are equal if and only if the signals f and g are equal.

5. (5 marks) Define the signals f and g as follows with $T_1 > T_2 > 0$,

$$f(t) = \frac{1}{T_1} \text{sinc}\left(\frac{t}{T_1}\right), \quad \text{and} \quad g(t) = \frac{1}{T_2} \text{sinc}\left(\frac{t}{T_2}\right).$$

Does the convolution $f * g$ exist? In other words, is the *signal* $(f * g)(t)$ well-defined for all values of t ?

If your answer is yes, first clearly explain why the convolution exists. And then derive $(f * g)(t)$.

If your answer is no, clearly explain why $(f * g)(t)$ does not exist for all values of t , i.e., identify at least one value of t where the convolution does not exist.
